

LATAM Bank Dispute Operations

An AI-assisted service system for one banking workflow: card-charge disputes, in Spanish and Portuguese.

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Live system

latam-bank-disputes.vercel.app

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THE PROBLEM, IN THE BANK'S DATA

Disputes stall before anyone can act

377

unrecognised-charge
complaints a month,
20% of all complaints

37

hours to a first answer
(median)

15

days to resolve (median)

16%

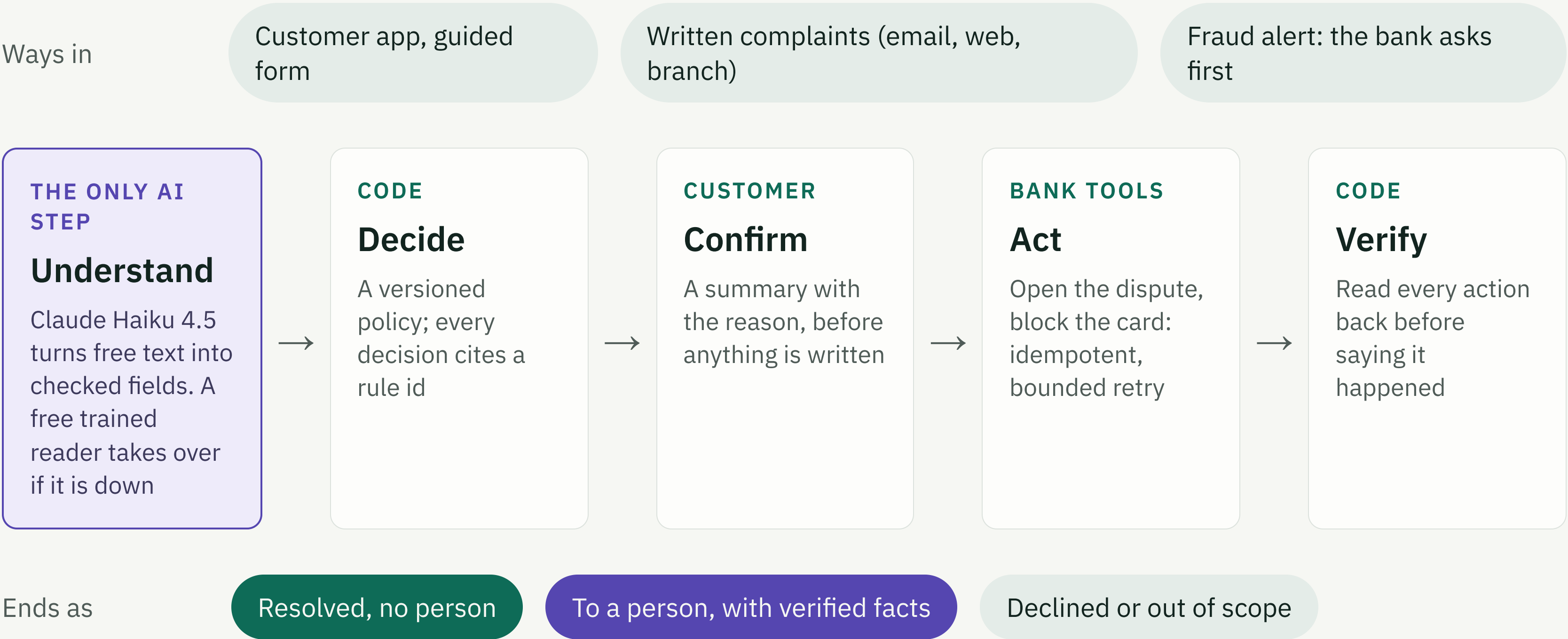
of written complaints
even identify the charge

The case breaks at the **intake**: the back office cannot act on what it was never told.
So the project covers one workflow, end to end, and automates the intake.

Organizer data (synthetic): 67,095 complaints, 2023–2026; "Cargo no reconocido" is the only sub-category of transaction complaints.

HOW IT WORKS

One case engine, three ways in



CONTROLLED AUTOMATION

Knowing when not to act

The model never writes

Opening a dispute or blocking a card is not a tool the model can call; the state machine calls it after policy and confirmation.

Identity comes from the session

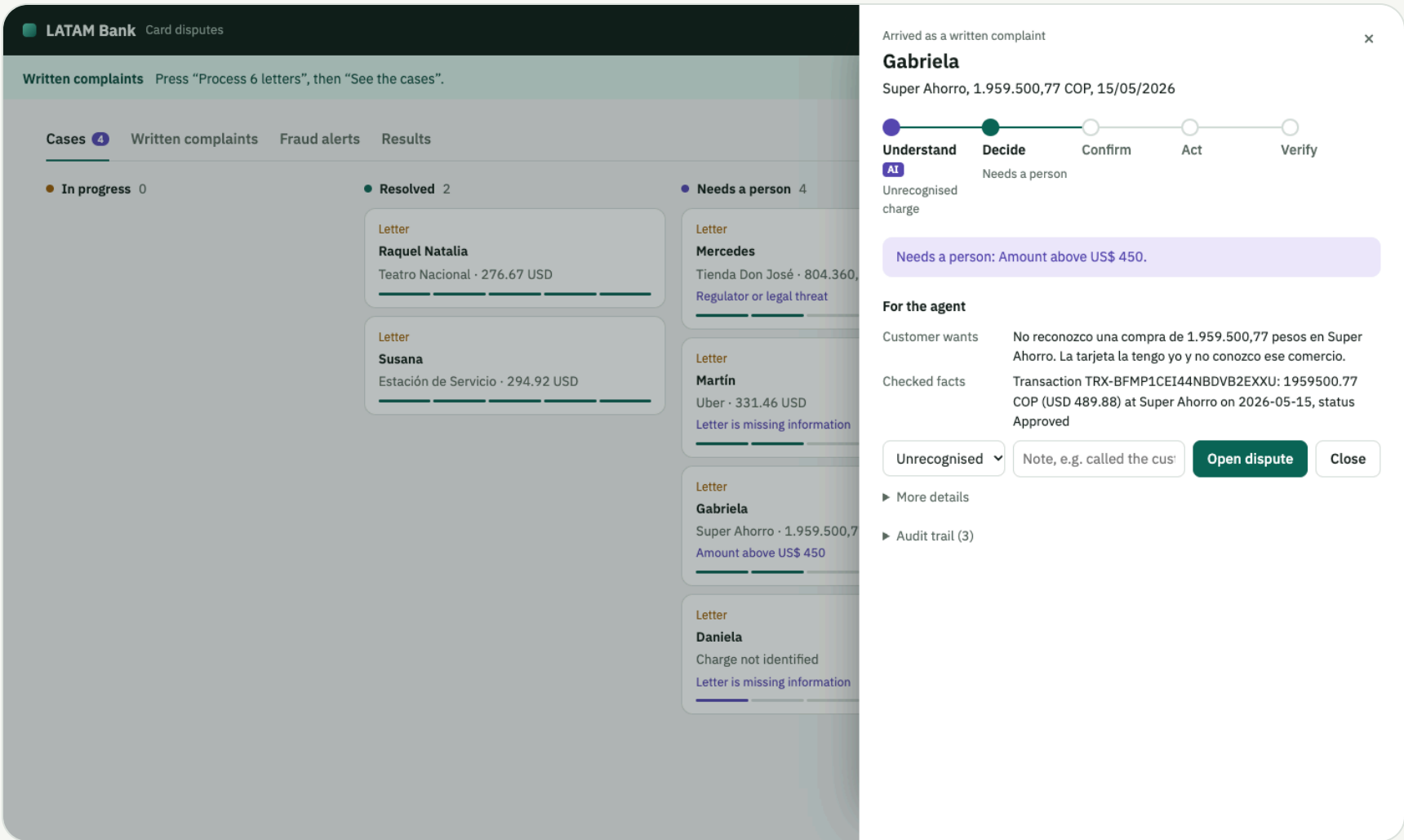
Tools resolve the customer from the token. "Not yours" and "does not exist" get the same answer.

A fact must have been said

On the real API the model guessed "card with me", and read a "not me" to a fraud alert as a yes. Now an unstated fact is asked, and the stated one outranks the model's intent.

Hand-offs a person can act on

Over US\$ 450, a regulator threat or a request for a person: verified facts and open questions, no transcript.



EVIDENCE

Measured on held-out cases

Held-out set	n	This system	Baseline
Vague memory, real Claude API	72	65 right · 4 unsafe	Chatbot 49 · 11 unsafe
Every reason, trigger and attack	84	84 right · 0 unsafe	Chatbot 56 · 13 unsafe
Written complaints	24	23 right · 0 unsafe	Keyword rules 19
Reason reading, blind labels	525	93.5%	Keyword rules 45.7%
Real bank calls sent to intake	872	7.9% (lower is better)	Earlier model 31.1%

3.1 s per turn (median) · US\$ 0.020 per safe resolution, chatbot US\$ 0.033 · 3 of the 4 unsafe outcomes fixed since and re-checked on the API (14 of 14)

Offline simulations: customers played by Claude Sonnet; outcomes expected by the written policy, judged from the database. The chatbot gets the same policy and tools. Not production.

UNDERNEATH

Data and ML, checked the same way

Data engineering

Bronze, silver, gold with data contracts.
The incremental run equals a full rebuild on 5.7M rows (4.5 s vs 12.1 s).
Gold is published only if conservation, volume and column-profile gates pass.

Machine learning

Ten targets in the bank's data tested: none worth a model (fraud AUC 0.50).
Trained instead: the ES/PT reason reader, 93.5% vs 45.7% for keywords; on 1,090 real calls, wrong routing 31% → 8%.

Analytics

Seven tested insights, each tied to a decision, including a correction: the "70% still open" backlog is a generator label, so the project stopped citing it.

What remains

A pilot that measures back-office minutes per case · the Claude reader on real speech · telling a forgotten subscription from fraud · Portuguese tested only on simulated and European speech.